



A new fallback beetle antennae search algorithm for path planning of mobile robots with collision-free capability

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Abstract

With the development of technology, mobile robots are becoming more and more common in industrial production and daily life. Various rules are set to ensure that mobile robots can move without collision. This paper proposes a novel intelligent optimization algorithm, named fallback beetle antennae search algorithm. Based on the analysis of biological habits, when the creature enters blind alley during the foraging process, it will retreat a distance and then restart the search process. We introduce a fallback mechanism in the traditional beetle antenna search algorithm. In addition, the proposed algorithm possesses the characteristic of low time complexity. It can plan a collision-free path in a short period of time. Moreover, the effectiveness and superiority of the algorithm are verified by simulations in different types of environments and comparisons with existing path planning algorithms.

Keywords Fallback beetle antennae search (FBAS) · Path planning · Obstacle avoidance · Mobile robots · Optimization

1 Introduction

In recent years, mobile robots have begun to play an important role in people's lives. More and more works are handed over to mobile robots (Guo and Zhang 2014; Chen and Zhang 2016, 2017; Li et al. 2017). In general, the research field of mobile robots can be divided into the following major parts, system design and hardware manufacturing, navigation and positioning and various applications (Guo and Zhang 2012; Chen and Zhang 2018; Xiao and Zhang 2014; Yang and Gao 2016; Chen et al. 2018b). Therefore, these research fields have also attracted the attention of many researchers (González et al. 2016; Ni et al. 2016; Zhang et al. 2016b; Chen et al. 2018a, c, 2019). For example, Mur-Artal and Tardós (2017) designed a tightly coupled visual-inertial simultaneous localization and mapping system to solve the scale blur

problem of monocular cameras. Gupta et al. (2017) proposed a target tracking solution for mobile robots based on online K–D tree-based classifier along with a Kalman filter. Rezaee and Abdollahi (2014) conducted a related study on the formation control of multiple mobile robots.

Navigation and positioning can be generally divided into the following sub-directions: positioning, establishing maps and path planning (Zaki et al. 2014; Pau et al. 2018; Xue et al. 2018; Xiao and Zhang 2016). In these research directions, the collision-free path planning of mobile robots is an important direction (Zhang et al. 2013, 2016a; Velagic et al. 2006; Sariff and Buniyamin 2006; Zhu and Yan 2010; Cong and Ponnambalam 2009; Zhong et al. 2016c). The task of collision-free path planning can be summarized as a collision-free path from the starting point to the ending point according to certain criteria in an environment with obstacles. According to the map information, it can be subdivided into two sub-directions. One global path planning (with the map information being known) and local path planning (with the map information being partly known or totally unknown) (Stentz A 1994; Stachniss and Burgard 2003; Luo and Yang 2008; Van et al. 2006; Rathbun et al. 2002; Chen et al. 2015). In the real world, it is impossible for a mobile robot to always work in an environment where obstacle information is known. Therefore, local path planning is more

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practical than global path planning. In this paper, we study the problem of local path planning.

In previous studies, researchers have used many different methods to solve the path planning problem of mobile robots considering collision-free conditions: for example, A-star algorithm (Bandi and Thalmann 2000), Dijkstra's algorithm (Dijkstra EW 1959), artificial potential field algorithm (APF) (Koren and Borenstein 1991) and ant colony algorithm (Mavrouniotis and Yang 2011), etc.

The A-star algorithm is a method commonly used in global path planning. An improved A-star algorithm named time-efficient A-star algorithm was proposed by Gurugi et al. (2016) to solve the path planning problem of mobile robots. The time complexity of the optimized A-star algorithm is still high. The APF is an algorithm commonly used for local path planning. It sets a gravitational field to the target point and a repulsive field to the obstacle. The object reaches the target point by the resultant force of the two fields, but can easily fall into the local extremum and cannot reach the target point. Lee and Park (2003) solve the problem that APF easily falls into local extremum by setting virtual obstacles and applying it to path planning. The ant colony algorithm is an intelligent algorithm. Liu et al. (2017) combined APF and geometric local optimization in the process of searching for the global optimal path. This algorithm can optimize the path through multiple iterations. As the problem size increases, it will take more time and become more likely to fall into local extremum.

Note that the above algorithms have deficiencies. We introduce the fallback mechanism based on BAS (Jiang and Li 2018), termed fallback beetle antennae search (FBAS), and applied the improved algorithm to the path planning of mobile robots. The proposed algorithm has the advantage of low time complexity while effectively planning a collision-free path. We proposed the FBAS algorithm and applied it to mobile robots path planning. The rest of this article is organized as follows. In Sect. 2, we describe in detail the problem definitions that need to be solved in this paper. In Sect. 3, we introduce the FBAS algorithm from two aspects: algorithm flow and improvement of the algorithm. In Sect. 4, the performance of the algorithm is evaluated through some simulations and comparisons with other algorithms. In Sect. 5, we summarize the full paper with the final remarks. The main contributions of this paper are as follows.

- By adding the fallback function to the conventional BAS strategy, a novel intelligent optimization algorithm, named FBAS, is proposed to improve the ability to avoid falling into local extremum.
- The proposed FBAS algorithm has the ability to avoid obstacles in the local path planning problem of mobile robots and has the advantage of fast path planning.

- The effectiveness of the algorithm is verified by the mobile robots applications. By comparing with other existing algorithms, the superiority on fast path planning for the mobile robot in consideration of obstacle avoidance is thus substantiated.

2 Problem formulation

This paper mainly solves the problem of local path planning. Specifically, it is necessary to solve such a problem: A collision-free path is planned in a static environment where the obstacle information is unknown. The map is set in advance, and the obstacles are also static obstacles set in advance, and the robot only knows the obstacle information in the current detection range. The starting point and the target point are also given in advance. The map is embodied in the program as a 0–1 matrix $M \in \mathbb{R}^{a \times a}$ depicted in the following form:

$$M_{i,j} = \begin{cases} 1, & \text{if } M_{i,j} \in O, \\ 0, & \text{if } M_{i,j} \notin O, \end{cases} \quad (1)$$

where $M_{i,j}$ is the ij th element in M and the O denotes obstacle area. The a represents the map size. Our proposed algorithm is suitable for path planning problems with objective functions, and the general form of the objective function with constraints in path planning can be described as

$$\begin{aligned} \min \quad & g(\xi) = (\xi - \delta)^2 \\ \text{s.t.} \quad & \xi \notin O \end{aligned} \quad (2)$$

where $g(\cdot): \mathbb{R}^k \rightarrow \mathbb{R}$ is the function to be optimized and $\xi \in \mathbb{R}^k$ represents coordinates of the current point. $\delta \in \mathbb{R}^k$ denotes coordinates of the target point. k is the data dimension which refers to the dimension of the planning space.

When all the path points in the path point array Q planned by the FBAS algorithm are not in the obstacle regions, we think that the path planned this time is effective. We find that the path is very tortuous, when we connect the path points directly. Therefore, a smoothing operation is required on the generated path to ensure that it can be used for navigation of mobile robots. While ensuring the feasibility of the path, the planning speed is also an important property. The faster the planning speed, the more practical the algorithm is. Based on the above description of the problem, the goal we need to achieve is clear: planning a smoother collision-free path between the starting point and the target point for the mobile robot as quickly as possible.

3 FBAS algorithm

The main purpose of the proposed algorithm is to plan a collision-free path from the starting point to the target point on a high-resolution map. Such an approach will get the next coordinate point in each iteration and get the path through multiple iterations. After smoothing the path, the final path is obtained. It should be noted that the map used in this paper is preset.

In order to understand the specific process of the FBAS algorithm, Fig. 1 shows a visual display of the algorithm, where black dot represents coordinates of the current point. Two small dots represent two candidate points. Blue dots represent the selected candidate point. Dotted circle indicates the current detection range. Brown rectangles represent obstacles area. The s_n and τ_n , respectively, represent the current step size and the current detection range radius. The n denotes the number of iterations.

The whole algorithm can be divided into three parts: candidate point generation, candidate point selection strategy and path generation. The three above-mentioned parts will be described in detail as follows.

3.1 Candidate point generation

The goal of the first part is to select two points around the current point as candidate points. The candidate points are determined in the planning space according to the random direction vector $\mathbf{d} \in \mathbb{R}^k$ and the detection range τ

$$\mathbf{d} = \frac{\text{rands}(k, 1)}{\|\text{rands}(k, 1)\|_2}, \quad (3)$$

where $\text{rands}(\cdot)$ function is a random number generation function, and k is the data dimension, which refers to the dimension of the planning space. In this paper, $k = 2$. After obtaining the direction vector $\mathbf{d}$, coordinates of two candidate points $\mathbf{x}_l \in \mathbb{R}^k$ and $\mathbf{x}_r \in \mathbb{R}^k$ are, respectively, generated

according to the detection range τ and the direction vector $\mathbf{d}$ obtained by Eq. (3)

$$\mathbf{x}_l = \mathbf{x} + \mathbf{d}\tau^i, \quad (4)$$

and

$$\mathbf{x}_r = \mathbf{x} - \mathbf{d}\tau^i, \quad (5)$$

where $\mathbf{x} \in \mathbb{R}^k$ is the coordinates of current point. i is the current number of iterations. τ^i denotes the current detection range, and it is calculated by the following equation:

$$\tau^i = \eta\tau^{i-1}, \quad (6)$$

where η represents the attenuation rate of the detection range. Adjusting η can improve the ability to avoid obstacles.

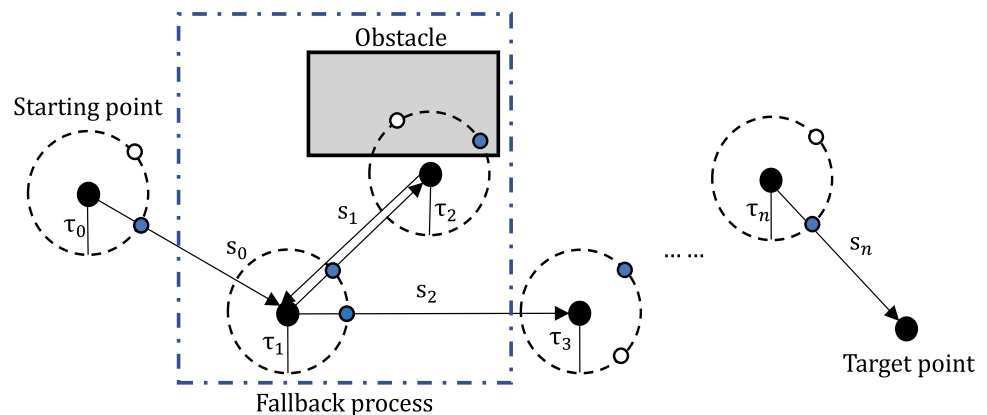
3.2 Candidate point selection strategy

The purpose of this part is to determine the coordinates of next path point based on the candidate point selection strategy. According to Eqs. (4) and (5), the coordinates of two candidate points have been obtained. FBAS algorithm is suitable for path planning problems with objective functions. An objective function is set in this paper, $f(\cdot) : \mathbb{R}^k \rightarrow \mathbb{R}$ which is essentially a function to be optimized. The next path point is determined based on the value of the objective function. The following defines the function to be optimized

$$f(\mathbf{x}) = (\mathbf{x} - \mathbf{x}_{\text{tar}})^2. \quad (7)$$

The $\mathbf{x}_{\text{tar}} \in \mathbb{R}^k$ is the coordinates of the target point. The $\mathbf{x}$ is the coordinates of the current point. In this paper, the smaller the objective function value, the closer the object is to the target point, so the overall purpose is to minimize the objective function. However, if the entire process minimizes the objective function, it may cause the mobile robot with high probability to fail to avoid non-convex obstacles.

Fig. 1 Block diagram of path planning process using the proposed FBAS algorithm



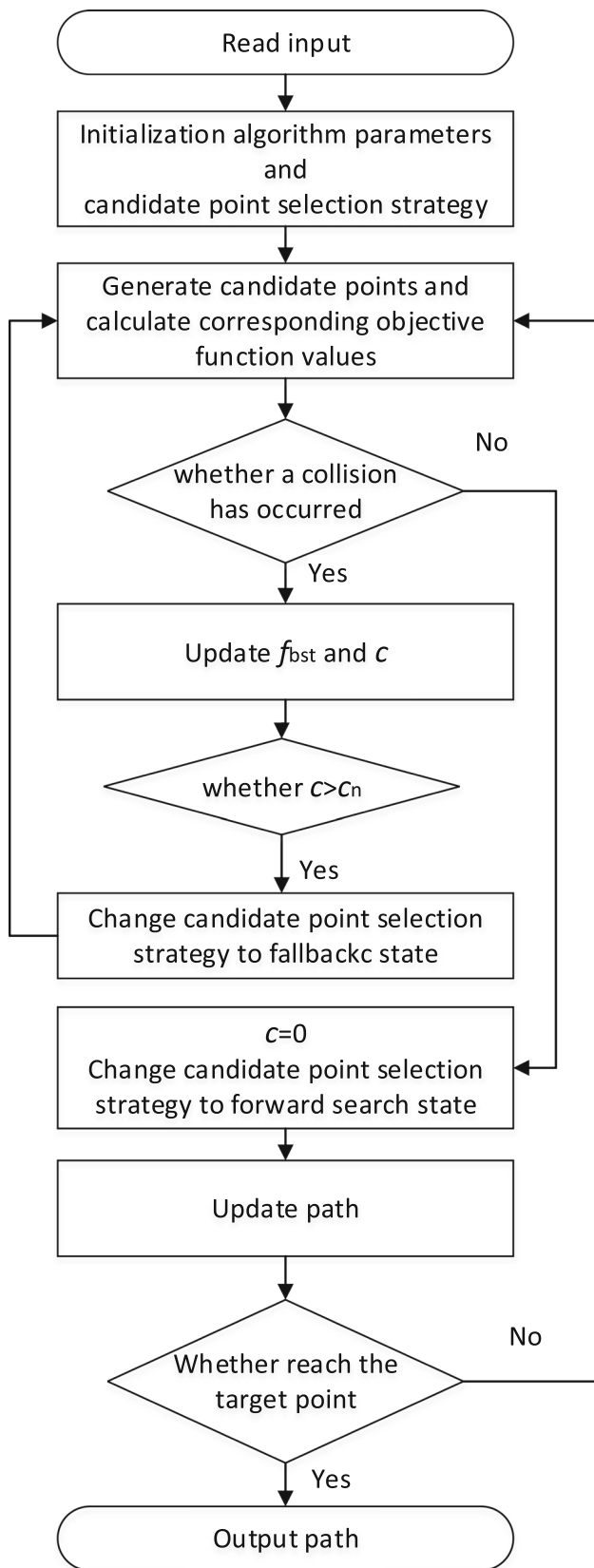


Fig. 2 Flowchart of the proposed FBAS algorithm

Algorithm 1: FBAS algorithm for collision-free path planning of mobile robots

Input: Two-dimensional map with obstacles M , the starting point $\mathbf{x}_{sta}$, $\mathbf{x}_{tar}$, the maximum number of iterations n , $f(\cdot)$;
Output: The path point array Q , the path length p ;

- 1 Initialize the basic parameters of FBAS such as $s, \tau, \varphi, \eta, l, f_{bst}, c, c_n$;
- 2 **for** $i = 1:n$ **do**
- 3 Randomly generated direction vector $\mathbf{d}$ via Eq. (3);
- 4 Select two candidate points $\mathbf{x}_l, \mathbf{x}_r$ via Eqs. (4) and (5);
- 5 Calculate the $f(\mathbf{x}_l)$ and $f(\mathbf{x}_r)$ via Eq. (7);
- 6 Randomly generated random walk direction vector $\mathbf{w}$ via Eq. (3);
- 7 Calculate $\mathbf{x}_{nex}$ based on selection strategy and Eq. (8);
- 8 Calculate the $f(\mathbf{x}_{nex})$ via Eq. (7);
- 9 **if** $\mathbf{x}_{nex}$ collides **then**
- 10 $c = c + 1$;
- 11 Update f_{bst} according to the via Eq. (11);
- 12 **if** $c > c_n$ **then**
- 13 Change the candidate point selection policy to the fallback state;
- 14 Continue;
- 15 **else**
- 16 $c = 0$;
- 17 Change the candidate point selection policy to the forward search state;
- 18 Save $\mathbf{x}_{nex}$ to Q ;
- 19 **if** $f(\cdot)$ has converged **then**
- 20 Break;
- 21 Calculate p based on Q ;
- 22 Return Q, p ;

Based on the analysis of biological habits, if the beetle enters blind alley during the foraging process, it will retreat for a distance and re-forward. Therefore, a fallback function has been added to the algorithm. During the fallback process, the objective function will be maximized briefly. Thence, the entire objective function optimization process can be roughly divided into two states: One is the minimization of the objective function in the forward search process and the other is the maximization of the objective function in the fallback process. Correspondingly, there are two candidate point selection strategies: One is that the forward search state takes the candidate point with a larger value of the objective function, and the other is the candidate point with the smaller value of the objective function selected in the fallback state. In order to achieve the switching between the two states, the setting of the number of collisions c is introduced in this paper. If the new available optimal objective function value f_{bst} is not obtained for this iteration, i.e., $f(\mathbf{x}_{nex}) > f_{bst}$ or $\mathbf{x}_{nex} \in O$, then it is determined that this beetle's movement has collided and c increases by one. When c is bigger than a preset collision threshold c_n , the selection strategy of the candidate points is changed to simulate the fallback during the foraging process. How to generate the next path point

$\mathbf{x}_{\text{nex}}$ is different in the two states. The relevant equations are as follows:

$$\mathbf{x}_{\text{nex}} = \begin{cases} x + \mathbf{d}s^i \text{sign}(f(\mathbf{x}_r) - f(\mathbf{x}_l)) + \mathbf{w}l, & \text{if } c \leq c_n, \\ x - \mathbf{d}s^i \text{sign}(f(\mathbf{x}_r) - f(\mathbf{x}_l)) + \mathbf{w}l, & \text{if } c > c_n, \end{cases} \quad (8)$$

where $\mathbf{x}_{\text{nex}}$ is the coordinates of next path point. The l is the step size of random walk. The direction of random walk $\mathbf{w}$ is generated according to Eq. (3). The s^i is the current step size and is determined by the following equation:

$$s^i = \varphi s^{i-1}, \quad (9)$$

where the φ denotes step attenuation rate

$$\text{sign}(x) = \begin{cases} 1, & x > 0, \\ 0, & x = 0, \\ -1, & x < 0, \end{cases} \quad (10)$$

where $\text{sign}(\cdot)$ is a symbolic function. By adjusting $\mathbf{w}$ and l , the ability of the algorithm to move away from local minima can be improved. The c can switch from the forward search state to the fallback state. The algorithm exits the fallback state only when the optimization gets a better objective function value, i.e., $f(\mathbf{x}_{\text{nex}}) < f_{\text{bst}}$ and $\mathbf{x}_{\text{nex}} \notin O$. In order to ensure that the beetle will continue to feed forward after returning a certain distance, it is necessary to add a penalty factor ω to f_{bst} . Taking a suitable value for ω based on s^i and $f(\cdot)$ ensures that in the fallback state, as the c increases, the growth rate of f_{bst} is greater than the growth rate of $f(\mathbf{x}_{\text{nex}})$. Therefore after returning a certain distance, the case of $f(\mathbf{x}_{\text{nex}}) < f_{\text{bst}}$ will appear, thereby enabling the algorithm to switch to the forward search state

$$f_{\text{bst}} = \frac{\omega c f_{\text{bst}}}{c_n}. \quad (11)$$

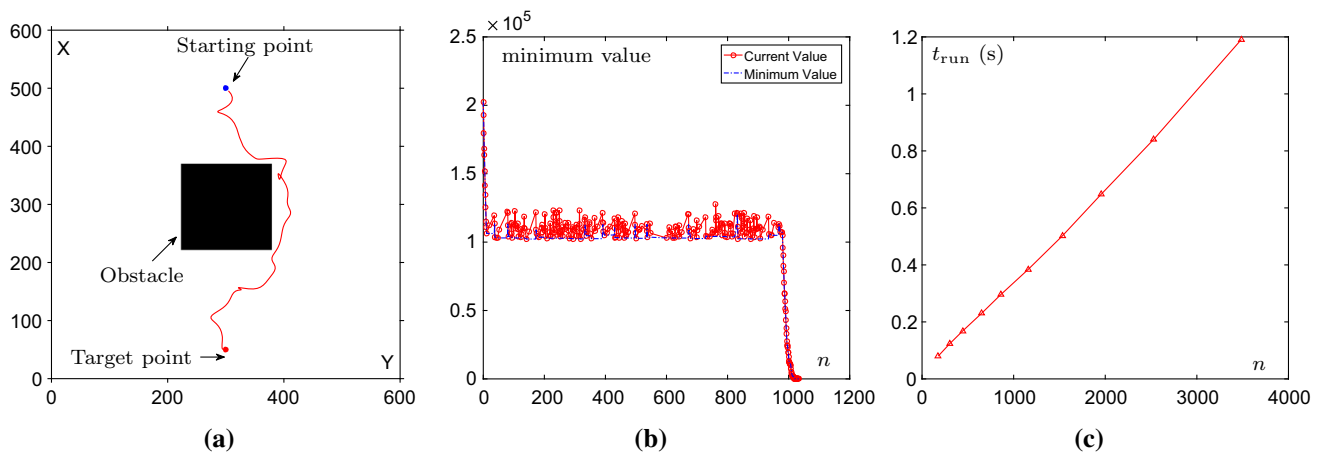


Fig. 3 Path planning results synthesized by FBAS algorithm to avoid single regular obstacle. **a** Motion results in a 2D plane. **b** Relationship between objective function and steps. **c** Relationship between run time and number of iterations

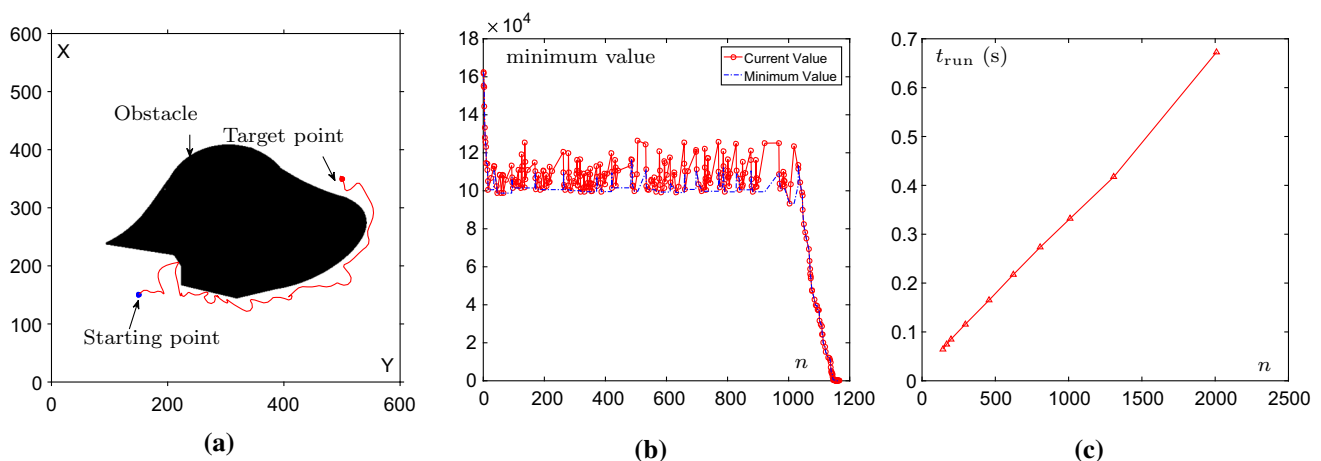


Fig. 4 Path planning results synthesized by FBAS algorithm to avoid single irregular obstacle. **a** Motion results in a 2D plane. **b** Relationship between objective function and steps. **c** Relationship between run time and number of iterations

Described above is the candidate point selection process in different situations. All path points can be obtained by repeating the contents of Sects. 3.1 and 3.2.

3.3 Path generation

The purpose of this part is to generate a collision-free path for mobile robots to use. Since the path points are already obtained in Sects. 3.1 and 3.2, this part directly performs the connection operation between the path points. The path generated by the direct connection path point is too tortuous, so we need to smooth the path. The B-spline curve is a common method for path smoothing in path planning (Tsai et al. 2011; Berglund et al. 2010; Elbanhawi et al. 2015; Koyuncu et al. 2008). The B-spline curves have such feature; a B-spline curve is contained in the convex hull of its control polyline (strong convex hull property). So the B-spline curve (Eilers and Marx 1996) can achieve the effect of modifying the local path without changing the shape of the entire path. $N_{i,q}(\cdot)$

are the normalized B-spline basis functions defined by the following Cox–de Boor recursion formulas:

$$N_{i,0}(u) = \begin{cases} 1, & \text{if } u_i \leq u < u_{i+1}, \\ 0, & \text{if otherwise,} \end{cases} \quad (12)$$

and

$$N_{i,q}(u) = \frac{u - u_i}{u_{i+q} - u_i} N_{i,q-1}(u) + \frac{u_{i+q+1} - u}{u_{i+q+1} - u_{i+1}} N_{i+1,q-1}(u), \quad (13)$$

where $N_{i,q}(\cdot)$ is the i th B-spline basis function of order q . The u_i is an element in an incrementing array $[u_0, u_1, \dots, u_{\max}]$. This paper exploits the B-spline curves for path smoothing. While $q = 3$, the relationship between the B-spline curve and the control point is as follows:

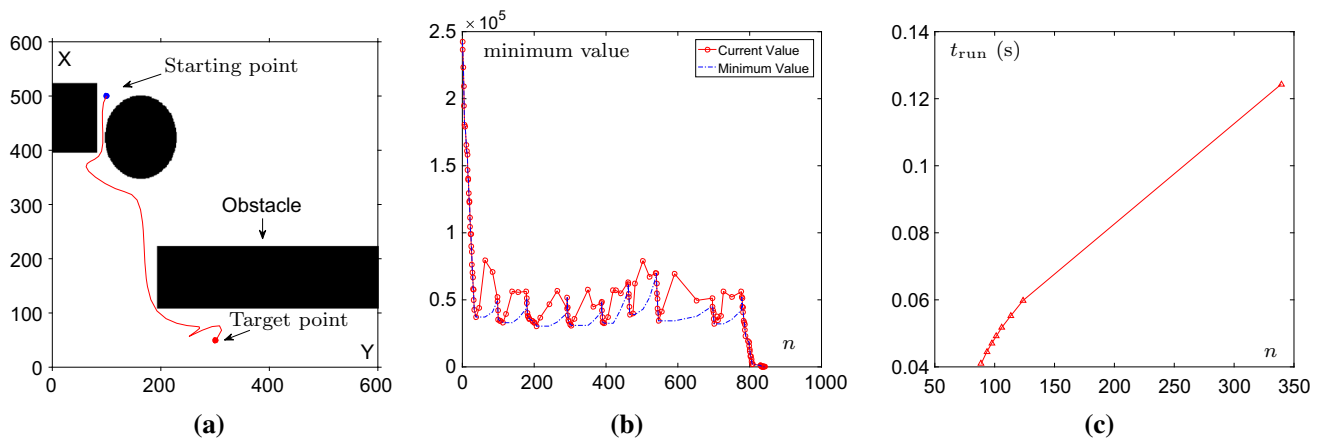


Fig. 5 Path planning results synthesized by FBAS algorithm to avoid multi-regular obstacle. **a** Motion results in a 2D plane. **b** Relationship between objective function and number of iterations. **c** Relationship between run time and number of iterations

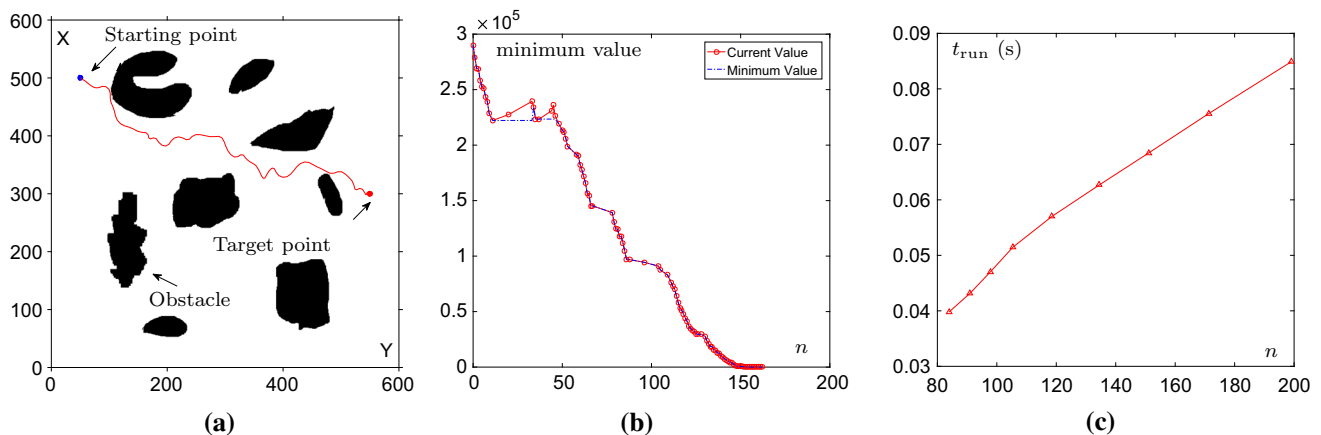


Fig. 6 Path planning results synthesized by FBAS algorithm to avoid multi-irregular obstacle. **a** Motion results in a 2D plane. **b** Relationship between objective function and number of iterations. **c** Relationship between run time and number of iterations

$$S(t) = \frac{1}{6} \begin{bmatrix} t^3 & t^2 & t & 1 \end{bmatrix} \begin{bmatrix} -1 & 3 & -3 & 1 \\ 3 & -6 & 3 & 0 \\ -3 & 0 & 3 & 0 \\ 1 & 4 & 1 & 0 \end{bmatrix} \begin{bmatrix} \mathbf{p}_i \\ \mathbf{p}_{i+1} \\ \mathbf{p}_{i+2} \\ \mathbf{p}_{i+3} \end{bmatrix}. \quad (14)$$

The resulting B-spline curve is defined by $S(\cdot)$. The $[\mathbf{p}_i, \mathbf{p}_{i+1}, \mathbf{p}_{i+2}, \mathbf{p}_{i+3}]^T$ is the set of control points. The numerical matrix and coefficients are fixed values that can be derived. After visualizing the generated path, a complete path planning process is over.

In Algorithm 1, we detailedly describe the main flow of FBAS algorithm and present it in the form of pseudocode.

In addition to help understand Algorithm 1, we present the flowchart of the entire algorithm in Fig. 2.

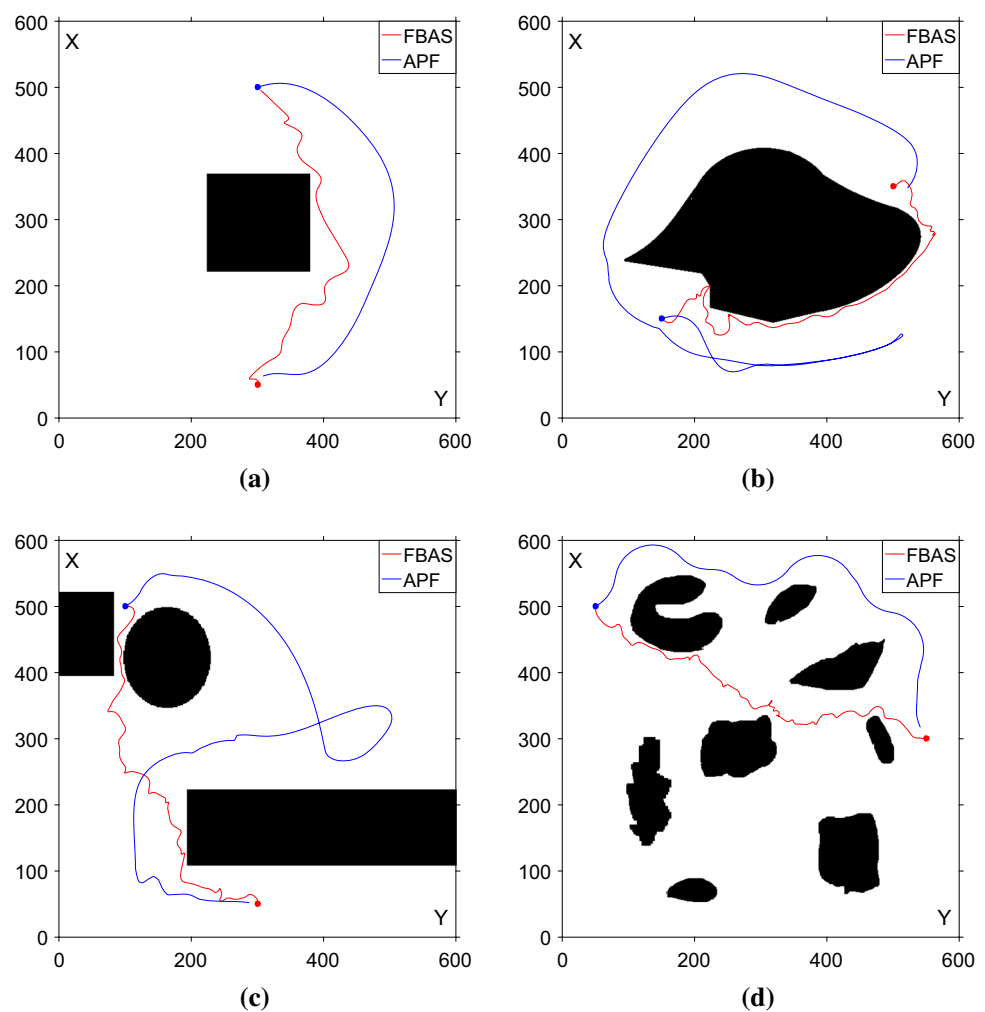
4 Simulations and comparisons

The proposed FBAS algorithm is suitable for path planning problems with objective functions. The purpose of this section is to verify the effectiveness of the FBAS algorithm by a series of simulations and comparison with other algorithms. We divided it into two subsections for analysis. We perform many simulations to verify the effectiveness of the algorithm. A volumeless point is used to simulate a mobile robot for

Table 1 Performance of FBAS algorithm for path planning to avoid different types of obstacles (with r denoting the success rate)

Type of obstacle	$\mathbf{x}_{\text{sta}}$	$\mathbf{x}_{\text{tar}}$	p	t_{run} (s)	n	r
Single regular	500,300	50,300	1.10×10^3	1.091	3187	1.00
Single irregular	150,150	350,500	1.01×10^3	0.618	1842	1.00
Multiple regular	500,100	50,300	8.83×10^2	0.654	1729	0.95
Multiple irregular	500,50	300,550	9.98×10^2	0.095	231	1.00

Fig. 7 Comparison results of path planning synthesized by the proposed FBAS algorithm and the APF on four different maps. **a** Single rule obstacle. **b** Single irregular obstacle. **c** Multiple regular obstacles. **d** Multiple irregular obstacles



reducing computation time. In Sect. 4.1, we selected four of the simulation results for in-depth analysis. This proves the robustness and versatility of the FBAS algorithm. In Sect. 4.2, we compare the FBAS algorithm with other algorithms. The comparison proves the superiority of our proposed algorithm.

All simulations in this paper are implemented on MATLAB, and all simulations use the same set of parameters. The specific parameter settings are as follows. The map size is 600×600 pixels. The step size $s = 15$. The sensing diameter $\tau = 7$. The step attenuation rate $\varphi = 0.9$. The sensing diameter rate of decay $\eta = 0.9999$. The random walk size $l = 1$. The threshold of the number of collisions $c_n = 200$. The penalty factor $\omega = 1$.

4.1 Illustrative examples

The purpose of this section is to verify the effectiveness of the algorithm through multiple simulation results. The simulation maps shown in this section can be divided into multiple obstacles and single obstacle according to the number of obstacles, or divided into regular obstacles and irregular obstacles according to the types of obstacles. In this paper, the simulation results are introduced separately according to multiple obstacles and single obstacles.

4.1.1 Single obstacle

First, Fig. 3 is a simulation result of avoiding a single regular obstacle. Subfigure (a) visualizes the path planning results. Subfigure (b) shows changes in the value of objective function with number of iterations. This result proves that our proposed algorithm has the ability to plan a collision-free path in the environment of a single regular obstacle.

Figure 4 is the simulation result of avoiding a single irregular obstacle. The difference between the obstacles in Figs. 3 and 4 is that one is a regular obstacle and the other is an irregular obstacle. Similarly, subfigure (a) is a visual representation of the simulation results for a single irregular obstacle. Sub-

Table 2 Comparisons of the proposed FBAS algorithm and the APF algorithm for path planning to avoid different types of obstacles

Type of obstacle	p		t_{run} (s)	
	FBAS	APF	FBAS	APF
Single regular	1.10×10^3	6.71×10^2	1.091	3.179
Single irregular	1.01×10^3	1.68×10^3	0.618	3.480
Multiple regular	8.83×10^2	1.26×10^3	0.654	3.280
Multiple irregular	9.98×10^2	1.01×10^3	0.095	5.178

Bold values indicate the significant advantages of FBAS in terms of planning time

figure (b) is the trend of objective function value with number of iterations. By combining the two simulation results, it can be concluded that the algorithm also has the ability to avoid obstacles for non-convex obstacles.

4.1.2 Multiple obstacles

Figures 5 and 6 are, respectively, simulation results for multiple regular obstacles and multiple irregular obstacles. Similar to the single-obstacle simulations, subfigure (a) intuitively shows the visual results of the path planning in the environment with multiple obstacles. Subfigure (b) shows the relationship between the objective function value and the number of iterations. Combining the above two simulation results, we can conclude that our proposed algorithm has the ability to avoid obstacles for multiple obstacles. By comprehensively analyzing the four simulation results in Sect. 4.1, we can prove that FBS has the ability to plan a collision-free path in environments with various obstacles.

4.1.3 Simulation result statistics

The purpose of this section is to evaluate the performance of the FBAS algorithm by the numerical results of the above four simulations. Table 1 is a statistic of the above four simulation results. The p is path length and n represents the number of

Fig. 8 Comparison results of the proposed FBAS algorithm and the APF on four different maps. **a** Running time. **b** Average path length

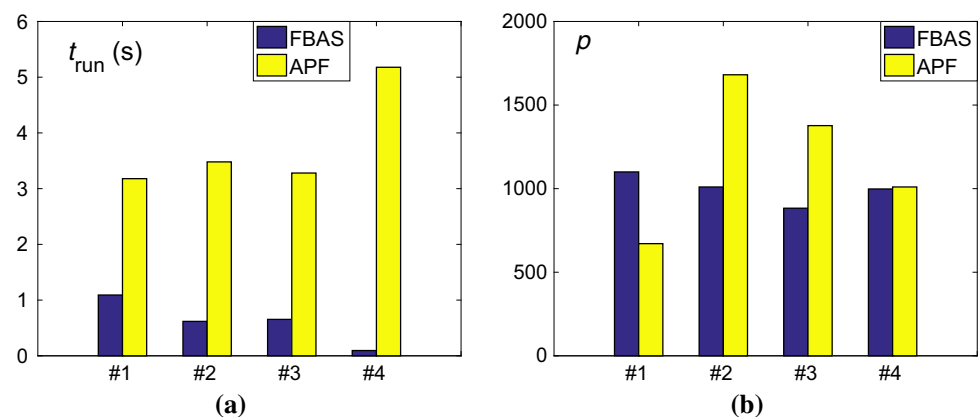


Table 3 Performance comparisons of the proposed FBAS algorithm and other algorithms

Algorithm	t_{run} (s)	p	r
FBAS algorithm	Short	Short	High
Algorithm A	Long	Long	Low
Algorithm B	Long	Short	High
Algorithm C	Long	Short	High
Algorithm D	Long	Short	Low
Algorithm E	Long	Short	Low

Algorithm A denotes the algorithm proposed by Koren and Borenstein (1991); Algorithm B denotes the algorithm proposed by Masehian and Sedighizadeh (2010); Algorithm C denotes the algorithm proposed by Guruji et al. (2016); Algorithm D denotes the algorithm proposed by Tan and Chen (2009); Algorithm E denotes the algorithm proposed by Montiel et al. (2015). Bold values indicate that FBAS has significant advantages in terms of planning time while other evaluation indicators and comparison algorithms are comparable

iterations. Each of the above maps is run 100 times using the same parameters, and the average path length and average

running time are obtained to evaluate the FBAS algorithm. As can be seen from Table 1, there are a higher success rate and shorter planning time on different maps.

4.2 Comparisons

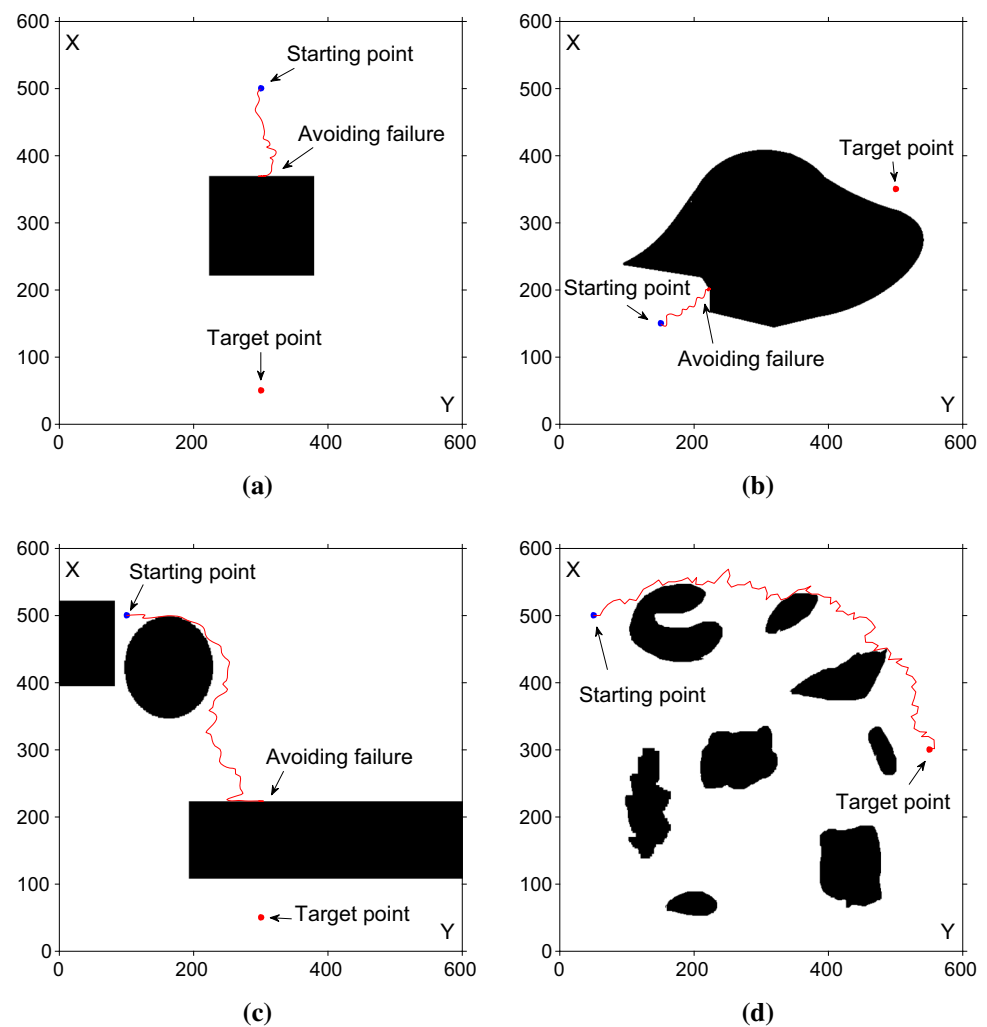
To further show the superiority of the proposed FBAS algorithm, the extensive comparisons with existing path planning algorithms are presented in this subsection.

4.2.1 Comparisons with APF

In order to further verify the effectiveness of the FBAS algorithm, we chose another algorithm commonly used for static path planning termed APF. APF and FBAS are compared in detail through a series of simulations. Figure 7 shows the results of path planning visualization of FBAS algorithm and APF on four simulation maps.

Since Fig. 7 is just a comparison of the results of a single simulation of the two algorithms. In order to avoid con-

Fig. 9 Comparison of path planning results by the conventional BAS algorithm on four different maps. **a** Single rule obstacle. **b** Single irregular obstacle. **c** Multiple regular obstacles. **d** Multiple irregular obstacles



tingency, we performed 100 simulations on each map in the same operating environment for the two algorithms and recorded the results in Table 2. It can be seen from the results that although the path obtained by the FBAS algorithm is not at all better than the APF in terms of path length, it has a significant advantage in terms of running time.

Figure 8 is a visual representation of the contents of Table 2, which provides a more intuitive understanding of the performance comparison between APF and FBAS. Combining Figs. 7 and 8, it can be seen that although FBAS algorithm has no obvious advantage in path length, it has a very significant advantage in running time.

4.2.2 Comparisons with other algorithms

In order to verify the effectiveness of the algorithm in many aspects, we make a simple comparison between FBAS algorithm and other algorithms. A total of three evaluation indicators were compared: running time, path length and success rate. The running time reflects the real-time extent of the

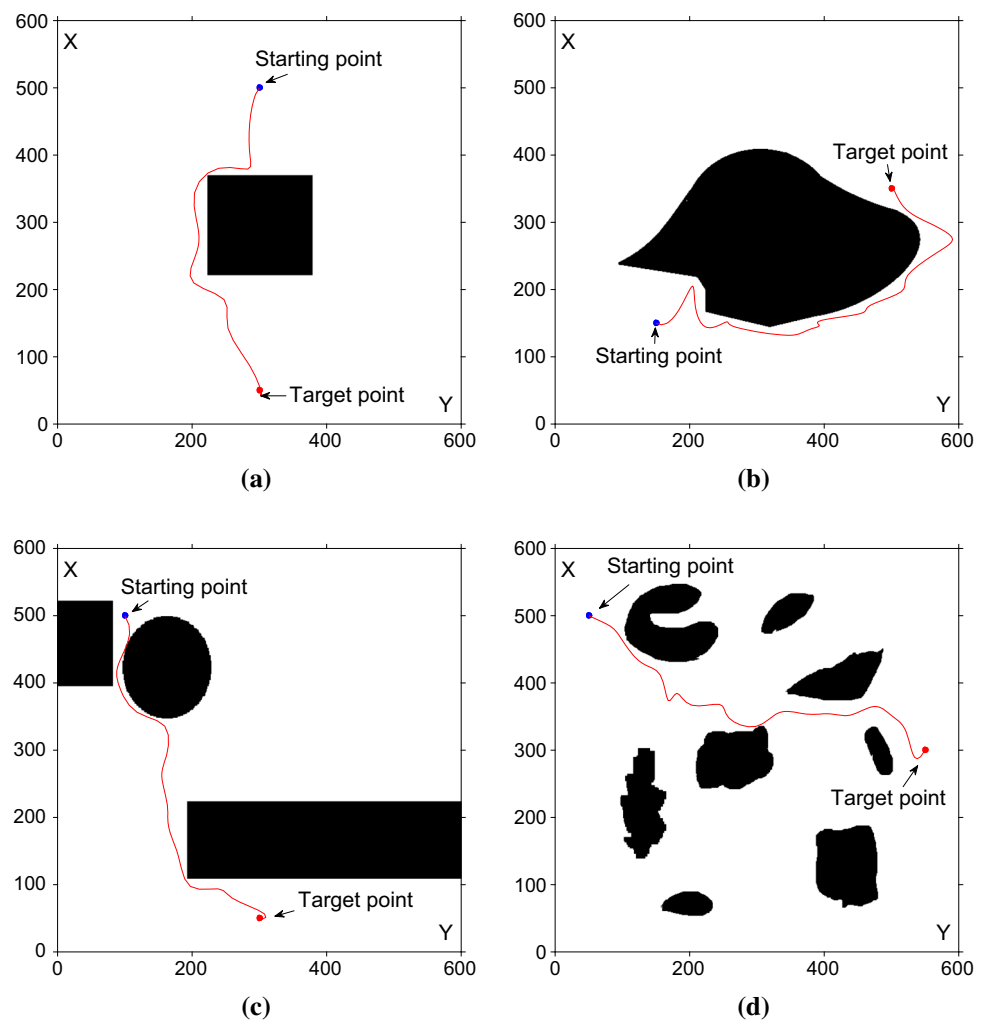
algorithm. The shorter the running time, the more likely it is to implement real-time path planning. Under the premise of successful obstacle avoidance, the path length is as short as possible. The success rate reflects the stability of the algorithm to some extent, and it can be said that it is not accidental to successfully avoid obstacles. The overall comparison results are shown in Table 3.

It can be seen from Table 3 that FBAS algorithm performs generally in terms of path length, but has significant advantages in terms of running time.

4.2.3 Comparisons with conventional BAS algorithm

In order to justify the fallback beetle strategy, we present a series of comparative results of FBAS with BAS in Fig. 9. It can be seen from Fig. 9 that BAS has the ability to easily avoid obstacles in the face of oversized obstacles and non-convex obstacles. In contrast, in Fig. 10, FBAS with the fallback function successfully reached the target point in various simulation maps. Through the comparison of the two

Fig. 10 Comparison path planning results by the proposed FBAS algorithm on four different maps. **a** Single rule obstacle. **b** Single irregular obstacle. **c** Multiple regular obstacles. **d** Multiple irregular obstacles



figures, the following conclusions can be drawn: The fallback mechanism has a significant effect on improving the ability of the BAS to avoid obstacles.

5 Conclusion

In this paper, we have proposed a new intelligent optimization algorithm based on BAS algorithm, termed FBAS. In order to verify the effectiveness of the algorithm, we have applied it to the path planning of mobile robots. The specific case is to plan a collision-free path in a static environment on the premise of a given starting point and ending point. In addition, we also have provided some simulation results to prove the effectiveness of the method. In the future work, we can replace static environments with dynamic environments and try to solve path planning in dynamic environments. We also hope that this research will encourage other scholars to explore this field in more depth.

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Compliance with ethical standards

Conflict of interest The authors declare that there is no conflict of interest in preparing this article.

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